



S2. Training & Validation Metrics

Accuracy tells us how often the model is correct. Loss tells us how wrong the model is.

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When we first learned Machine Learning, we spent a lot of time discussing evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. These metrics are still important, especially for classification problems.

However, once we move into Deep Learning, our attention shifts toward a different set of metrics. During training, we spend most of our time monitoring four values:

- Training Accuracy
- Validation Accuracy
- Training Loss
- Validation Loss

These four metrics tell us whether the model is learning, memorizing, underfitting, or overfitting.

Interestingly, although accuracy is often the metric beginners focus on, experienced practitioners usually pay much closer attention to the loss curves. The reason is simple: loss provides much more detailed information about the learning process, while accuracy only tells us whether a prediction is correct or incorrect.

For this reason, whenever we train a neural network, one of the first things we do is plot the training and validation loss curves and study their behavior carefully.

1. Training & Validation Accuracy

Accuracy measures the percentage of predictions that the model gets correct.

For example, if our model correctly predicts 900 out of 1000 samples, the accuracy is:

$$Accuracy = \frac{900}{1000} = 90\%$$

During Deep Learning training, we usually track two types of accuracy:

1.1. Training Accuracy

Training Accuracy measures how well the model performs on the training dataset.

As training progresses, we generally expect this value to increase because the model is becoming better at recognizing patterns from the data.

A very high training accuracy usually indicates that the model has learned the training set well.

1.2. Validation Accuracy

Validation Accuracy measures how well the model performs on unseen validation data. This metric is much more important because it tells us whether the model can generalize beyond the data it has already seen.

A model that performs well on training data but poorly on validation data is usually overfitting.

For example:

Epoch	Training Accuracy	Validation Accuracy
1	62%	60%
10	89%	87%
20	98%	78%

At first glance, the training accuracy looks impressive.

However, the validation accuracy starts dropping. This suggests that the model is beginning to memorize the training data rather than learning patterns that generalize to new examples.

Accuracy is therefore useful for understanding overall prediction performance, but it has a major limitation: ***It only tells us whether a prediction is right or wrong. It does not tell us how wrong a prediction is.***

To understand that, we need another metric.

2. Training & Validation Loss

Loss measures the error made by the model. Unlike accuracy, loss does not simply classify predictions as correct or incorrect. Instead, it measures how far the prediction is from the actual answer.

Because of this, loss provides a much more detailed view of the learning process.

2.1. Training Loss

Training Loss measures the error on the training dataset. During successful learning, we expect training loss to decrease steadily.

A decreasing training loss means that the neural network is adjusting its weights in the correct direction and gradually improving its predictions.

2.2. Validation Loss

Validation Loss measures the error on unseen validation data. This metric is one of the most important indicators in Deep Learning because it often reveals problems long before accuracy does.

Consider the following example:

Epoch	Training Loss	Validation Loss
1	1.25	1.30
10	0.55	0.60
20	0.20	0.75

Notice what happens. **The training loss continues decreasing**, which means the model is becoming increasingly specialized to the training data. **However, the validation loss starts increasing.** This is one of the clearest signs of overfitting.

The model is still improving on data it has already seen, but it is becoming worse at handling new data. This is why we often say: ***The moment validation loss starts increasing while training loss continues decreasing, overfitting has begun.***

3. Why Do We Care More About Loss Than Accuracy?

Suppose we have two models:

Model	Accuracy
A	90%
B	90%

Both models appear identical.

However, their losses may look very different:

Model	Loss
A	0.10
B	0.80

The accuracy cannot distinguish between them.

The loss can.

Model A is making confident and mostly correct predictions.

Model B is making many uncertain predictions and may be very close to making mistakes.

In other words:

- Accuracy measures correctness.
- Loss measures confidence and error magnitude.

Because neural networks learn by minimizing loss rather than maximizing accuracy, loss gives us a much clearer picture of what is happening during training.

This is why, whenever we examine training curves, we usually focus on the loss curves first and use accuracy as a secondary indicator.

A model with healthy loss curves is usually learning properly. A model with unhealthy loss curves is often heading toward underfitting or overfitting, even if its accuracy still appears good.

4. Overfitting and Underfitting

At this point, we know that Deep Learning models are trained by minimizing a loss function. During training, we continuously monitor two values: Training Loss and Validation Loss

These two curves tell us whether the model is learning properly, struggling to learn, or beginning to memorize the training data.

While accuracy is useful, loss is usually the first metric we examine because it provides a much clearer picture of what is happening inside the training process.

Let us explore the most common scenarios.

4.1. Scenario 1: Both Training Loss and Validation Loss Increase

This is usually the worst situation.



As training progresses, both losses become larger and larger. Instead of learning useful patterns, the model is moving further away from the correct answers.

This often happens when:

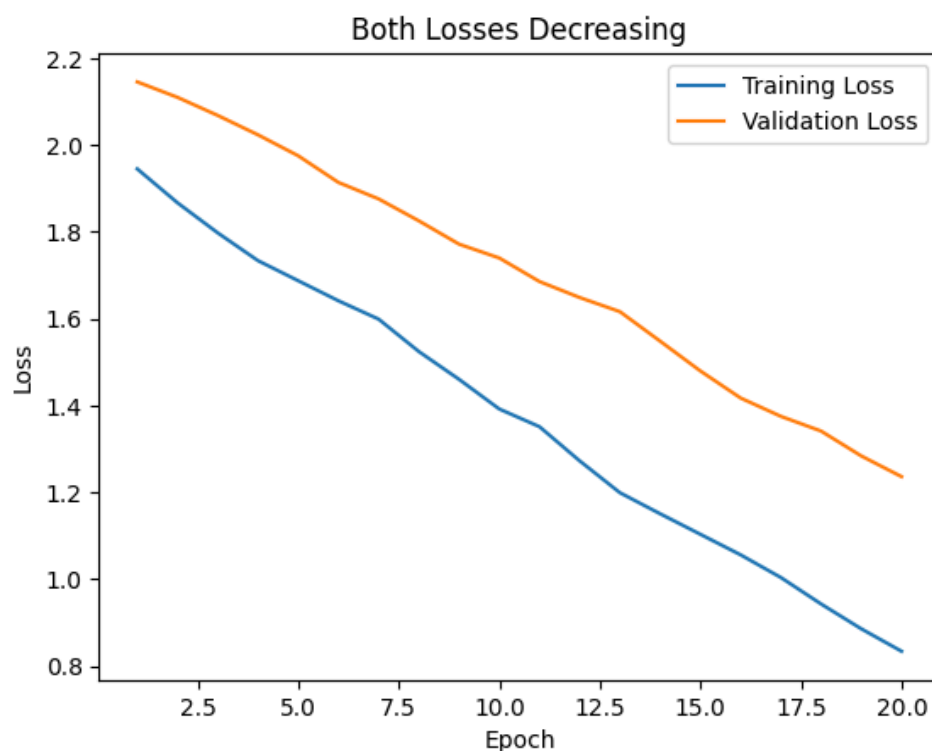
- The learning rate is too high
- The model architecture is unsuitable
- The data contains severe problems
- Training becomes unstable

In simple terms, the model is not learning. If we continue training, the performance will likely become even worse.

This is a clear sign of failed training and severe underfitting.

4.2. Scenario 2: Both Training Loss and Validation Loss Decrease

This is exactly what we want to see.



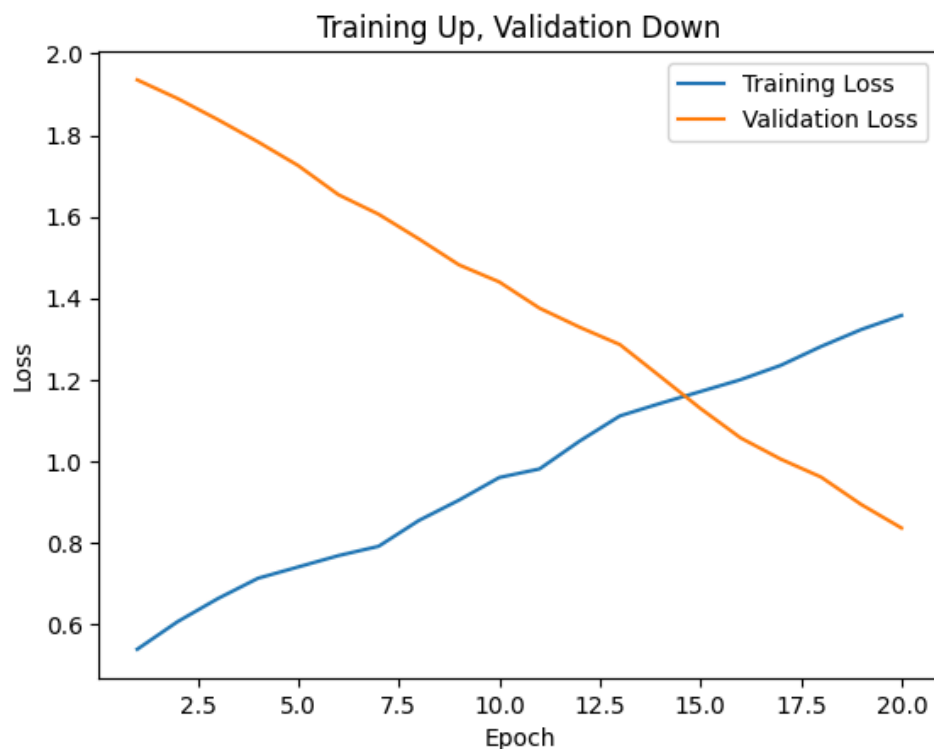
As training progresses: **Training Loss decreases** and **Validation Loss decreases**

The model is improving on the training data while also improving on unseen data. This indicates that the model is successfully discovering meaningful patterns rather than memorizing examples. The closer these two curves remain to each other, the better the model is generalizing.

When beginners ask what a healthy training process looks like, this is usually the answer.

4.3. Scenario 3: Training Loss Increases while Validation Loss Decreases

This situation is quite unusual. The model appears to perform worse on the training data while somehow improving on the validation data.



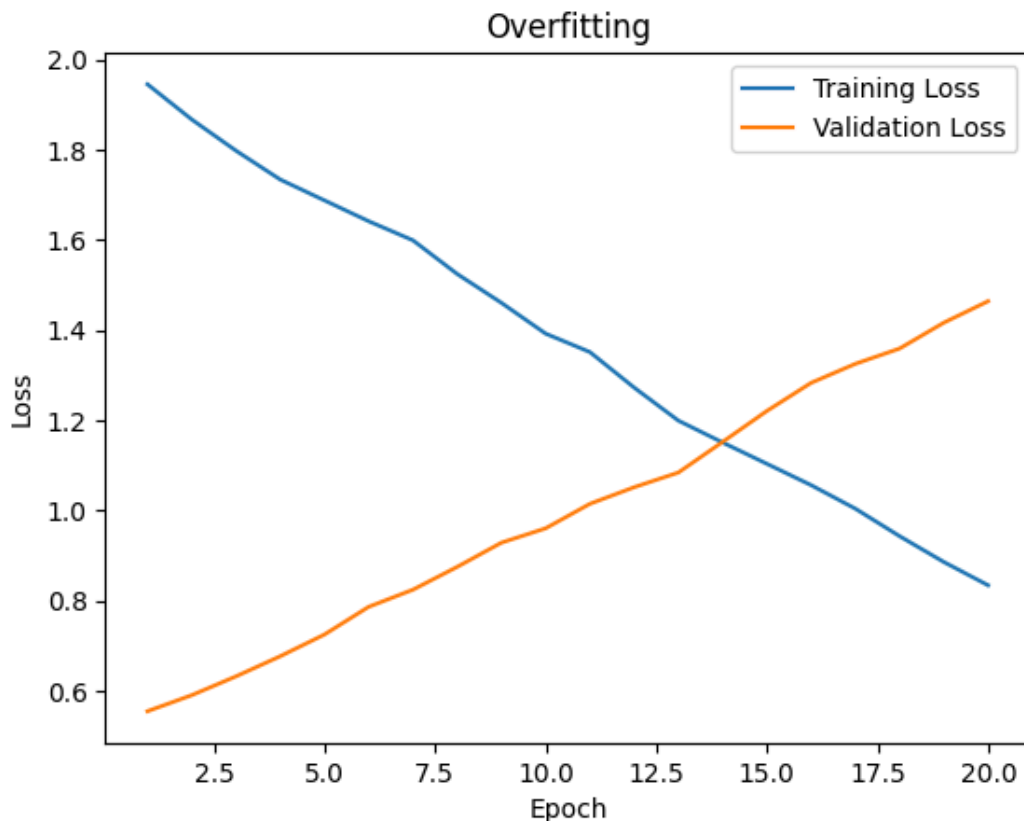
In practice, this is rarely caused by genuine learning behavior. More commonly, it occurs because of excessive regularization, heavy dropout, noisy training batches, measurement inconsistencies

Although the validation loss may look promising, the increasing training loss suggests that the model is struggling to fit the data.

This often indicates instability rather than successful learning.

4.4. Scenario 4: Training Loss Decreases while Validation Loss Increases

This is the classic sign of overfitting.



The model continues improving on the training data since the Training Loss decreases. However, its performance on unseen data becomes worse, as Validation Loss increases

At first, the model learns meaningful patterns. Eventually, it begins memorizing details specific to the training set. The model becomes increasingly specialized to the training data and loses its ability to generalize.

This is why many practitioners monitor the validation loss very carefully. **The moment validation loss starts increasing while training loss continues decreasing, overfitting has begun.**

In many projects, the best model is not the one from the final epoch. Instead, it is often the model saved just before the validation loss starts rising. This idea forms the foundation of Early Stopping.

The goal is not to make the training loss as small as possible. The goal is to make both losses decrease together and remain close to one another. A model that learns useful patterns will improve on both training and validation data. A model that only improves on training data is simply memorizing.

Summary

This section covers one of the most important concepts in Deep Learning:

Understanding learning behavior through loss (or accuracy) curves.

By observing the relationship between Training Loss and Validation Loss, we can identify successful learning, failed learning, underfitting, and overfitting.

More importantly, these curves help us decide when to stop training, adjust hyperparameters, or redesign the model before poor generalization becomes a serious problem.

References

<https://www.geeksforgeeks.org/deep-learning/training-and-validation-loss-in-deep-learning/>

Next Section:

 [S3. Learning Rate](#)